

Effective, Specific, and Non-Selective: Activation Interventions on Sequentially Computed State

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Abstract

Interpretability research increasingly validates activation interventions by showing that they produce an intended counterfactual answer. RAVEL established that this is insufficient for static entity attributes: a faithful intervention must also leave unrelated attributes intact, and MIB reports that full-vector interventions fail exactly this test. We ask whether the same distinction holds when the target is not a stored attribute but a state computed over several reasoning steps. Using a transfer task with a known causal program ($\text{holder}_{t+1} = \text{recipient}_t$), we construct an intervention that passes a progressively strengthened battery of controls: it changes the answer, it is specific to the donor whose state was transplanted ($\text{DS} = +0.718$, with non-injected donors at 0.000), it is inert under a norm-matched random direction at every magnitude, and it is graded in magnitude. We then query the same intervened forward pass about quantities the causal program says must not change. Selectivity is 0/240 in two model families (95% CI [0.0, 1.6]%). Phi-3.5-mini answers “which object moved?” with 100% accuracy unintervened, and with a person’s name afterwards. Behavioral control of a reasoning answer does not imply control of the reasoning state.

1 Introduction

Activation interventions are usually validated by a single output: patch a representation, observe the counterfactually correct answer, conclude the representation carried the variable of interest. Interchange-intervention accuracy formalizes this as output agreement with a high-level causal model (Geiger et al., 2021).

RAVEL (Huang et al., 2024) showed this is only half the requirement, at least for static entity attributes: intervening on a city’s continent should change the continent answer *and leave the language answer alone*. MIB (Mueller et al., 2025)

builds this into a benchmark and reports that full-vector baselines do poorly on RAVEL precisely because swapping a whole vector disturbs unrelated attributes.

Both concern attributes the model *retrieves* — a city’s continent is static, present because the entity was mentioned. We ask whether the same distinction holds when the target is a state the model must *compute*: a value in no single input token, defined only after several reasoning steps.

Our contribution is an extension, not a new method or a new diagnostic: RAVEL’s effectiveness/selectivity distinction survives the move from static attributes to sequentially computed state, and an intervention passing a control battery substantially stronger than “the answer changed” still fails it completely.

We do not claim selective interventions on reasoning state are impossible, that interchange-intervention accuracy is invalid, or that existing methods choose sites carelessly — DAS (Geiger et al., 2024) restricts NLI interventions to [CLS]; MIB brute-forces manually selected locations. This is a counterexample in a controlled setting, bounded accordingly in Section 6.

Relation to activation steering. A parallel literature adds learned directions to hidden states to control behavior, and validates them behaviorally. Our result bears directly on that practice: a direction can produce graded, specific, norm-controlled behavioral change while leaving the intended semantic variable untouched, so behavioral validation alone does not license a semantic interpretation of the direction.

2 Task: a sequentially computed causal variable

We use a transfer task in which objects pass between people:

Ava has the ring. Liam has the book.
Ava gives the ring to Emma.
Liam gives the book to Henry.
Emma gives the ring to Owen.
Henry gives the book to Grace.
Who has the ring? → Owen

The causal program is explicit:

holder₀ = initial holder in the first sentence
holder_{t+1} = recipient_t
 Z_{curr} = holder_k after k transfers

Three properties make this a usable testbed.

The target is computed, not retrieved Z_{curr} is not a property of any entity in the prompt; it is the result of following a chain. A recency heuristic fails by construction, because each round ends with a *distractor* transfer, so the most recently named person never holds the queried object.

One body, several variables Over one unchanged chain we can query the current holder, the last recipient, the original holder (holder₀), the identity of the transferred object, and the number of transfers. The causal program says an intervention setting Z_{curr} must change the first two and must not change the last three. This yields a *predicted consequence pattern* rather than a single expected answer.

Vocabulary is controlled Names and objects are drawn from pools filtered so every answer is a single token in both tokenizers, checked against the in-context form rather than a naive leading-space form, which SentencePiece splits. Donor and receiver items use disjoint names and objects, so a successfully transported answer is a token absent from the receiver’s own context and cannot be explained by its surface content.

3 Intervention protocol

Let h_D and h_R be the residual-stream activations of a donor and a receiver at layers $L = [18, 20, 22, 24]$, at the final pre-answer token position. The intervention adds

$$v = \alpha \cdot (h_D - h_R)$$

to the receiver’s activations at those layers and that position, with $\alpha = 1$ unless stated. At $\alpha = 1$ this

quantity	baseline	after $do(D)$
$P(Y_{\text{donor}})$, injected	0.000	0.718
$P(Y_{\text{other}})$, not injected	0.066	0.000
$P(Y_{\text{receiver}})$, own answer	0.604	0.000

Table 1: Donor specificity as a cross-term within one patched forward pass.

is exact replacement of the receiver’s state with the donor’s.

Position The final pre-answer token was selected empirically before any held-out item was scored, and the choice is consequential: at the mid-sentence recipient-name position the same oracle intervention scored 0/8, while at the final position it scored 6/8 with the identical layer set. A position sweep (Section 5.5) finds effects *only* at the final position.

Few-shot prefix Every prompt carries a few-shot prefix covering all five question forms, so no probe is disadvantaged by unfamiliar formatting and priming is identical across conditions. This matters: with a two-form prefix the current-holder transport rate was 24/24; under the five-form prefix it is 74.2%. We report the harder number.

Donor question The donor prompt always ends with “Who has the {object}?”, so its final-token state corresponds to $do(Z_{\text{curr}} := z'_D)$. That single state is then patched into receivers asked *different* questions. This is what makes the consequence pattern measurable.

4 The control stack

Before testing selectivity we establish that the intervention is worth taking seriously. Each control is progressively harder to pass.

Effectiveness. The patched receiver produces the donor’s answer. On Phi-3.5-mini, 74.2% [65.7, 81.2] on the current-holder probe.

Donor specificity. Measured as a cross-term within a single patched pass, not by comparing two separate interventions (Table 1). $DS_{\text{cross}} = +0.718$. Top-1 is the injected donor 13/16; top-1 is a non-injected donor 0/16. The output tracks which donor was supplied and does not drift toward arbitrary names.

probe	question	change?	probe	chg?	baseline	→don	→rec
current_holder	Who has the {obj}?	YES	curr_holder	Y	99.2	74.2	0/120
last_recipient	Who received it last?	YES	last_recip	Y	97.5	69.2	0/120
original_holder	Who originally had it?	NO	orig_holder	N	100.0	60.0	0/120
object_identity	What object did X have?	NO	obj_identity	N	100.0	71.7	0/120
transfer_count	How many times given away?	NO	xfer_count	N	20.8*	85.8	0/120

Table 2: The five probes over one unchanged chain body. YES means the probe must take the donor’s value; NO means it must retain the receiver’s.

A measurement correction worth recording.

We initially compared “patch $D \rightarrow P(Y_D)$ ” against “patch a random donor $R \rightarrow P(Y_R)$ ”, found both succeeded at similar rates (75% vs 72%), and read this as absence of specificity. That inference is wrong: both are legitimate interchange interventions with different donors, and the causal model predicts both should succeed. A random donor is not a control that ought to fail. Specificity must be a cross-term within one pass. We also verified that the near-equality was not driven by label collision (2/32 items had a random donor whose answer coincided with the correct donor’s; enforcing a collision-free derangement left the result unchanged) and that a true no-op self-patch reproduces the baseline exactly 32/32.

Norm control. A Gaussian direction rescaled to *exactly* the norm of $h_D - h_R$ produces a change in $P(Y_{\text{donor}})$ of **0.000 at every α** . The effect requires a genuine model-state direction; injected magnitude alone does nothing.

Dose response. $P(Y_{\text{donor}})$ rises smoothly with α — 0.005, 0.201, 0.601, 0.653, 0.718 at $\alpha = 0.125, 0.25, 0.5, 0.75, 1.0$ — with a threshold near $\alpha \approx 0.25$. The effect is graded, not an unstable all-or-nothing flip.

5 Selectivity evaluation

The donor state is patched into receivers asked each of five probes. The causal program predicts which must change (Table 2).

Competence gate. A selectivity failure is only interpretable on probes the model can answer *unintended*. Probes below 50% baseline competence are excluded, because a post-intervention change on a question the model could never answer tells us nothing.

Sampling. 3 seeds $\times$ 40 pairs = **120 independent pairs per model**. Each seed resamples both

Table 3: Phi-3.5-mini, percentages. *excluded by the competence gate. Completeness 172/240 = 71.7% [65.7, 77.0]; selectivity 0/240 = **0.0%** [0.0, 1.6]. Per-seed selectivity: 0/40, 0/40, 0/40.

probe	chg?	baseline	→don	→rec
curr_holder	Y	46.7*	50.0	0/120
last_recip	Y	73.3	44.2	0/120
orig_holder	N	100.0	46.7	0/120
obj_identity	N	100.0	49.2	0/120
xfer_count	N	18.3*	49.2	0/120

Table 4: Llama-3.2-3B, percentages. *excluded by the competence gate. Completeness 53/120 = 44.2% [35.6, 53.1]; selectivity 0/240 = **0.0%** [0.0, 1.6].

the item pairs *and* the few-shot prefix. The intervention itself is deterministic — zero-initialized, eval() mode, no optimizer — so pairs and prefix are the only stochastic components, and the prefix is a genuine confound worth varying. Wilson 95% intervals are computed over independent pairs within each probe, never over pooled probe-outcomes.

5.1 Phi-3.5-mini (primary)

5.2 Llama-3.2-3B (corroborates selectivity only)

Llama’s completeness is *not* comparable to Phi’s, and we do not present it as such. Llama graded three probes; Phi graded four, because Llama’s current-holder probe fell below the competence gate at 46.7%. Llama cannot reliably perform the primary task at this sample size, so it supports the *selectivity* claim — where both its graded should-not-change probes have 100% baseline competence — and not the completeness claim.

5.3 Is the metric capable of firing?

A counter that always reads zero is indistinguishable from a broken counter, so we verify the instrument directly. The selectivity check and the baseline competence check are *the same comparison against the same target token*, differing only in whether the patch is applied. So the 100% baseline competence on original_holder and object_identity already demonstrates that

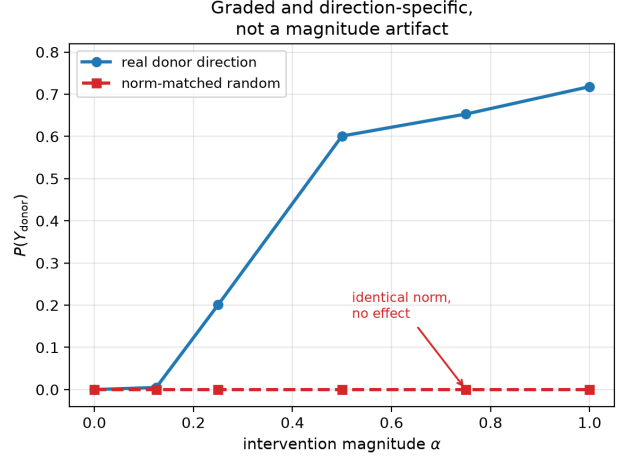
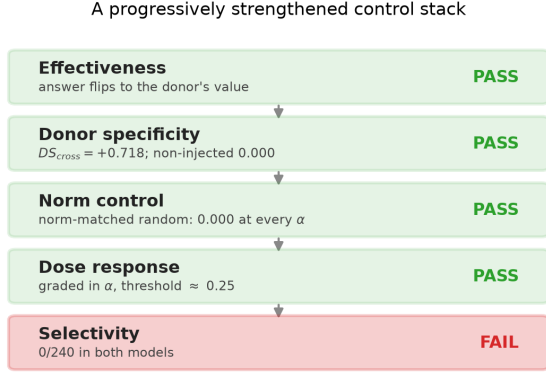


Figure 1: Left: each credential in the control stack, in the order applied. Right: the intervention is graded in magnitude, while a direction with identical norm produces no effect at any magnitude.

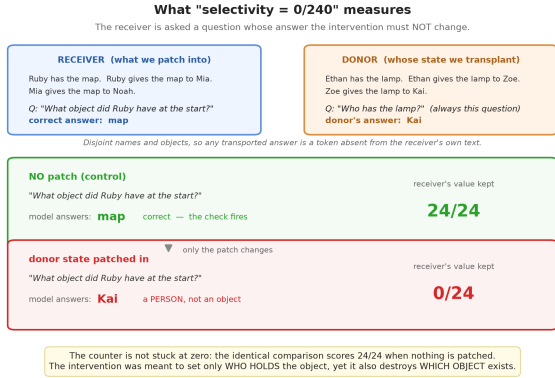


Figure 2: What the selectivity counter counts. The identical comparison scores 24/24 with no intervention and 0/24 with the donor patch; only the patch differs, so the zero reflects the intervention rather than a metric that cannot fire.

the selectivity check fires at ceiling when nothing is patched. We confirm this empirically in a single process on identical items ($n = 40$ per probe, both models):

model	zero patch	donor patch
Phi-3.5-mini	80/80	0/80
Llama-3.2-3B	80/80	0/80

Table 5: Only the patch differs. The metric is live in both models, and the intervention alone drives it to zero.

The same runs show where the probability mass goes: the model answers with the donor's value on 21/40 and 26/40 items (Phi), 19/40 and 21/40 (Llama), for original_holder and object_identity respectively.

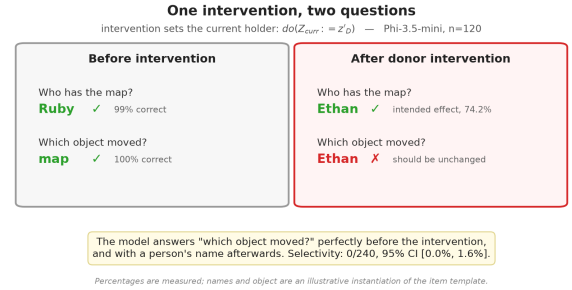


Figure 3: One intervention, two questions. The object-identity question is answered perfectly unintervened and with a person's name afterwards. Percentages are measured; names and object are an illustrative instantiation of the item template.

5.4 The concrete failure

Both models answer "What object did {first} have at the start?" with **100% accuracy unintervened**. After an intervention that, by construction, sets only who currently holds the object, Phi answers with a **person's name** on 71.7% of items. No reading of $do(Z_{curr} := z'_D)$ permits changing which object exists in the problem.

This is difficult to attribute to misunderstanding the question: the model answered it perfectly moments earlier. The intervention destroyed the semantic isolation between the targeted answer state and unrelated information.

5.5 Supporting observations

Three observations support the interpretation without being independent contributions, reported in full in Section B: the effect is confined to the answer boundary (0/12 transport at positions −2

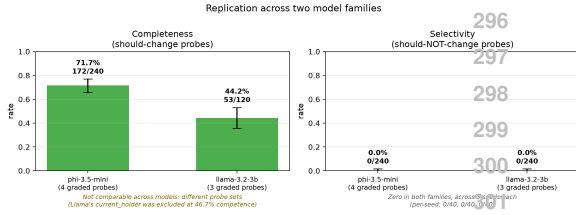


Figure 4: Replication across two model families. Wilson 95% intervals over independent pairs. The completeness bars are *not* comparable across models: Llama graded three probes to Phi’s four, because Llama’s current-holder probe fell below the competence gate.

through -7 versus $9/12$ at -1); compact subspaces do not carry it ($0/32$ at rank 32 for every basis tested, against $24/32$ full-space); and successful edits are not geometrically diverse, so intervention non-uniqueness does not explain the failure.

6 Discussion

What the experiment establishes. In this setting, an intervention can satisfy effectiveness, donor specificity, norm control, and dose-response while exercising no selective control over the reasoning variable it appears to manipulate. The four credentials are jointly insufficient. A two-probe consequence check detects this at negligible cost.

What it does not establish. It does not show that interchange-intervention accuracy is invalid — MIB uses it deliberately to align features with *specified* variables, and Sutter et al. (2025) already demonstrate a sharper limitation, obtaining 100% IIA on randomly initialized language models with sufficiently powerful alignment maps. Nor does it show that selective interventions on reasoning state are impossible; we tested one intervention family at one site.

The obvious objection, conceded. One may respond: you patched the final pre-answer token, so of course you manipulated an answer representation. We agree, and the position sweep is evidence for that reading — effects appear only where answer information becomes directly actionable. The point is not that a poorly chosen site gives a poor result. It is that an intervention at this site passes four escalating credentials a practitioner would reasonably accept. The credential stack, not the site, is the object of study.

Relation to prior work. RAVEL contributes the effectiveness/selectivity distinction; MIB con-

tributes its benchmark operationalization and the observation that full-vector interventions fail it for entity attributes. Our addition is narrow: the same failure occurs for a *computed* state, and it survives a control battery considerably stronger than answer-change alone.

7 Limitations

- One task family.** A single transfer-chain task with one causal program. We make no claim of universality.
- Selectivity rests on two probes per model.** `transfer_count` was excluded in both models (20.8% and 18.3% competence): neither can count transfers unprompted, so post-intervention changes there are uninterpretable.
- Llama supports only part of the claim.** Its current-holder competence is 46.7% at $n = 120$, so its completeness figure is computed on a different probe set and is not comparable to Phi’s.
- One intervention site and one layer set.** Layers [18, 20, 22, 24] at the final pre-answer position. Other sites may behave differently — indeed Section B shows they behave *very* differently.
- Competence is not shown to drive the effect.** Phi has both higher baseline accuracy and stronger intervention takeover, but two models cannot establish a relationship between competence and selectivity failure. The defensible statement is narrower: the selectivity failure cannot be explained by poor baseline task competence, since it persists in Phi at 99–100% baseline accuracy on all graded probes.
- Small-sample estimates were optimistic.** At $n = 24$ Llama’s current-holder competence appeared to be 88%; at $n = 120$ with freshly resampled items it is 46.7%. Early-stage numbers in this literature, including our own, should be treated with caution.

8 Future work

The sharp dependence on intervention position — full effect at the final token, nothing one token earlier — suggests that selectivity may vary systematically along the computational trajectory. Mapping

where interventions transition from state-level control to answer-level control, as a grid over layers and positions with a selectivity panel at each cell, is the natural next study and a substantially larger undertaking than the present one.

References

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A Probe templates and the competence gate

Item construction. Each item is a single transfer-chain body (Section 2) rendered once and then queried five ways, so completeness and selectivity are measured on the *same* underlying problem rather than resampled items per probe. The donor is always asked the canonical question; only the receiver is asked all five. The templated slots are filled from the receiver’s own problem, never the donor’s, so a probe answered correctly with the *donor’s* value cannot be explained by surface leakage from the question text itself.

Why the competence gate exists. A selectivity failure — a should-not-change probe moving after the patch — is only evidence of a semantic leak if the model could answer that probe correctly *unpatched*. If a model cannot perform the task at all, a post-intervention “change” on that probe is noise, not leakage. The gate is applied per probe, per model, at a 50% baseline-accuracy floor, before any selectivity number is computed or reported.

Why transfer_count was excluded for both models. Baseline accuracy was 20.8% (Phi) and 18.3% (Llama) — both models fail to count transfers unprompted at a rate indistinguishable from guessing among small integers, independent of any intervention. Reporting a “selectivity” result on this probe would conflate task incapacity with semantic leakage, so it is dropped from the verdict in both models, leaving four graded probes for Phi and three for Llama.

Answer-token construction. Count answers are rendered as words (“two”, not “2”), because the digit form tokenizes inconsistently across architectures — verified to silently corrupt generated pairs before being caught. All answers are checked for single-token status *in the rendered context*, not a naive leading-space heuristic, since SentencePiece-family tokenizers (Phi) split some such forms differently.

B Supporting observations

The effect is confined to the answer boundary. Patching at positions −2 through −7 (the queried object, “the”, “has”, “Who”, the sentence-final period, the recipient name) produces 0/12 transport for both correct and random donors, while position −1 produces 9/12. There is no intermediate regime in which the effect is present but weaker.

Compact subspaces do not carry it either. At rank 32, projected interchange through tracking, PCA, random, gradient-coordinate and learned-causal bases all produce 0/32, while the full-space oracle produces 24/32. Random bases fail at every rank in both architectures, and PCA performs comparably to a clean-minus-corrupt tracking basis through the useful rank regime — evidence that the tracking contrast does not identify a privileged reasoning subspace.

Successful edits are not geometrically diverse. We tested whether behaviorally equivalent edits

must be internally similar, since a negative answer would have offered an alternative explanation for our results. Across 17 restarts $\times$ 3 examples with randomized initialization, median pairwise cosine between successful edits is +0.840, +0.912, +0.948. Solutions travel $\sim 2.5\times$ their initialization norm and end nearly orthogonal to their starting points ($\cos \approx 0.02$), then converge on the same direction regardless. Independently initialized optimizations find essentially one solution, so intervention non-uniqueness does not explain the selectivity failure. This also accounts for the low cosine ($\approx 0.08\text{--}0.15$) observed between unconstrained and subspace-confined edits: that is forced by confining a vector to a rank-32 slice of a 3072-dimensional space, not evidence of distinct solutions.

C Reproduction

All experiments are frozen at a single snapshot; no number in this paper may be changed without a corresponding change to that snapshot. A verification script re-checks every quoted number against the frozen logs and fails the build on any mismatch (44 checks). Item generation, intervention mechanics, the cross-question panel, donor specificity, the magnitude and position sweeps, the uniqueness falsification, the selectivity-metric validation, and all figure generation are each a separate script in the release. Code and frozen logs will be released upon acceptance.

The panel runs predate an output flag added later and cannot be regenerated bit-for-bit, since each seed resamples both pairs and prefix; the figure data is a hand transcription of those logs, and figures are built from it.